Dominion Energy Virginia reliability: what public data shows, and where I'd look first

Independent analysis of public data

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November 3, 2026

Data: EIA-861 · PUDL · NOAA Storm Events · EIA-930

Outage minutes rose from 130 to 133; 2025 had more storms and more interruptions

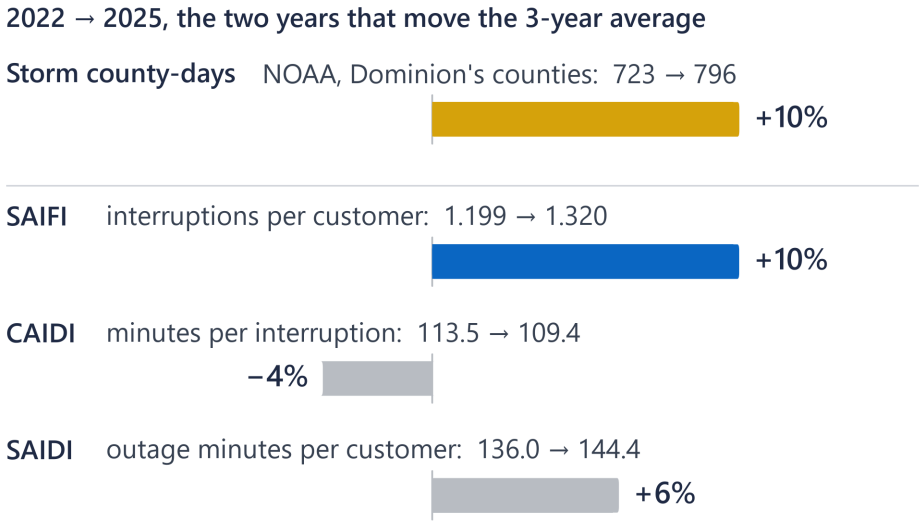
130 → 133

Reported 3-year average outage minutes per customer, excluding major events

Rebuilt from EIA-861: 130.2 → 133.0

2025 (144.4) replaced 2022 (136.0) in the window.

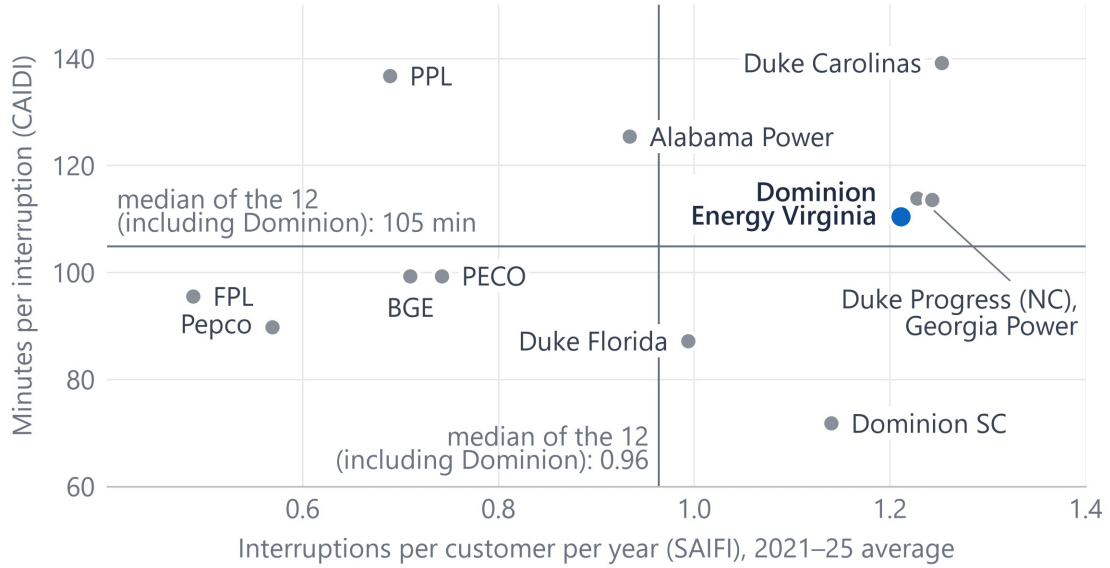
Dominion attributes the rise primarily to increased storm activity (2025 Annual Report, p. 10). 2025 sits on the storm line, with no clear trend once storms are counted (A2).



Why it matters: a stormy year can move a 3-year average. With outage records, I'd ask whether non-storm interruptions are rising underneath.

Source: EIA-861 2025 final and PUDL (EIA-861) 2013–2024, Dominion Energy Virginia, excluding major event days; 3-year means of annual values. NOAA Storm Events (A2). Reported figures: Dominion Energy 2025 Annual Report, p. 10.

Among 12 large utilities, Dominion's interruption length is mid-pack; frequency is the gap



1.21

interruptions per customer a year: 4th highest of the 12 (median 0.96)

110 min

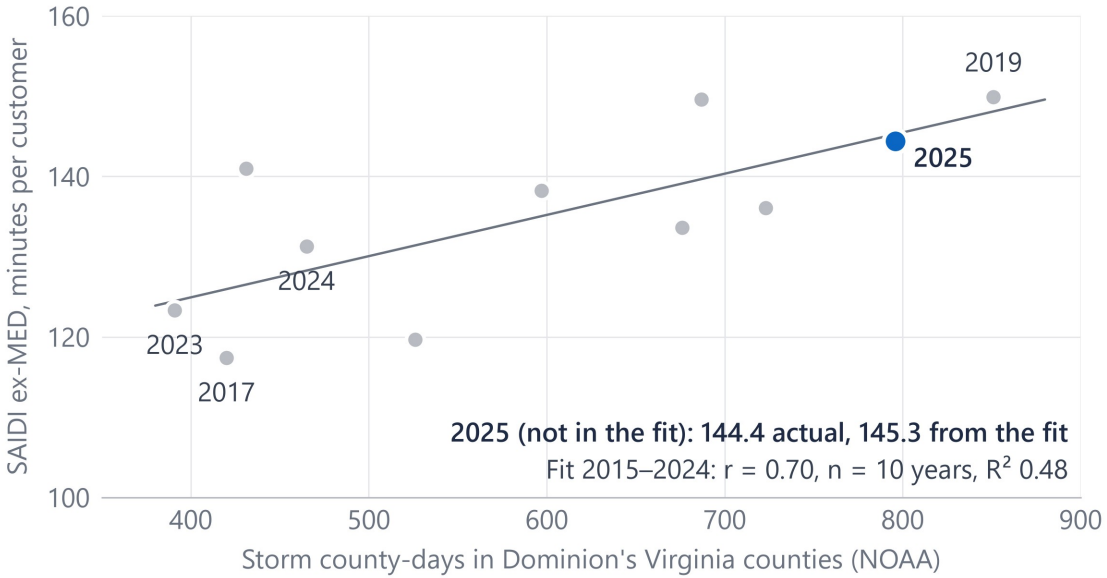
per interruption: mid-pack, 6th of the 12 (median 105)

Methods differ: Dominion reports under its own major-event rule; ranks are indicative.

Why it matters: if frequency is where Dominion differs, the next question is which causes drive interruptions, and that needs outage records.

Source: EIA-861 / PUDL 2021–2025, excluding major event days, means of annual values. Dominion and 11 peers: twelve large investor-owned utilities in PJM and the Southeast, selected by EIA ID before results were computed; medians include Dominion.

Years with more storm days had more outage minutes (r = 0.70 over ten years)



465 → 796

storm county-days in Dominion's Virginia counties, 2024 → 2025

r = 0.70

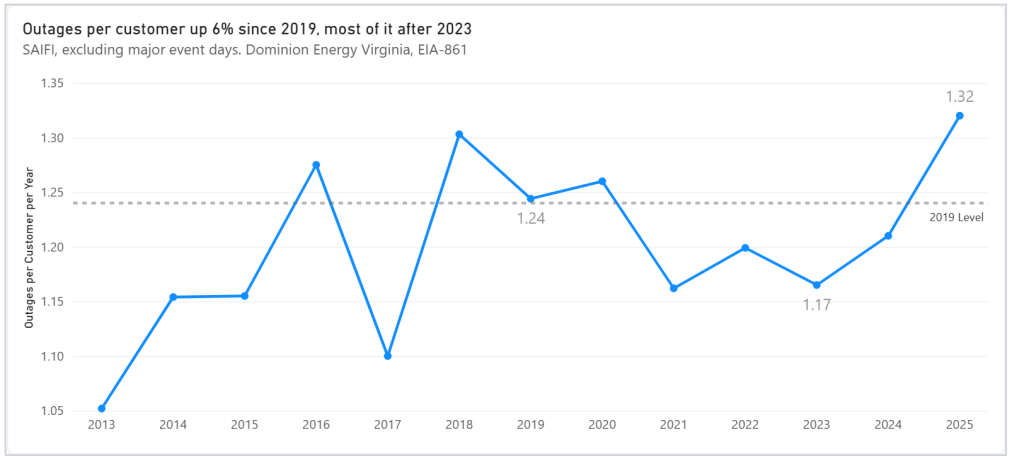
over 2015–2024 (n = 10): an association, not a cause. The fit, made without 2025, gives 145.3; the actual was 144.4.

Why it matters: reporting reliability next to storm exposure would keep a stormy year from being read as a trend.

Source: NOAA Storm Events 2015–2025, wind, winter and tropical event types, by county in Dominion's Virginia service territory (EIA-861); EIA-861 SAIDI excluding major event days; least-squares fit on 2015–2024.

The Power BI report: one page per question, on customer-weighted measures

My current report: the What moved page, frequency chart (EIA-861, Import mode)



FCT_RELIABILITY
utility × state × year

many:1

DIM_UTILITY
peer group

A small star at the core; the other tables are summaries keyed by year or hour. Import for presentation, DirectQuery during development (D13).

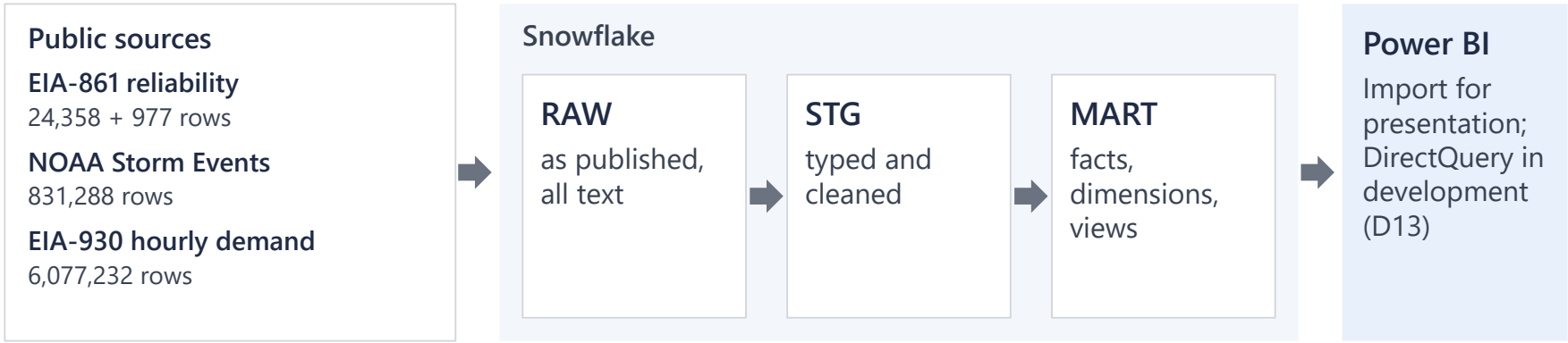
Customer-weighted measures; the 3-year figure averages the annual values:

```
SAIDI 3yr ending 2025 =
CALCULATE(
    AVERAGEX(VALUES(FCT_RELIABILITY[YEAR]),
        [SAIDI ex MED]),
    FCT_RELIABILITY[YEAR] IN {2023, 2024, 2025})
```

Why it matters: Power BI reporting leads the role's duties, and the same model and measures could carry a program-tracking report.

Source: Power BI Desktop, the project's report (.pbix; not on GitHub, by design): What moved page, Import mode. Measures as in the project's powerbi/measures.dax.

Built with production habits: tested, documented, and reconciled to the reported numbers



12 tests pass

keys, SAIDI = SAIFI × CAIDI, EIA's published 2025 values

Data dictionary

every table and column; measures defined in docs/05

Decisions log

decisions logged (D1–D14); the peer set was chosen by EIA ID before results

Reconciled

130.2 and 133.0 rebuilt against 130 and 133; 2025 matches EIA (test T4)

Why it matters: the same habits would serve the team's reports: reconcile first, test every load, write each definition down.

Source: Project build: SQL scripts, tests T1–T12, data dictionary and decisions log D1–D14; row counts as loaded.

With Dominion's outage, meter and circuit data: the questions I'd bring and what I'd build

Days 1–30

Learn and reconcile

Learn how outages, meter events and circuits are recorded and reported.

Rebuild one year's SAIDI, SAIFI and CAIDI and reconcile them to the filed figures.

Deliverable: a reconciled baseline with written definitions

Days 31–60

Ask about frequency

By circuit and cause code, storm days apart from calm days; repeat interruptions (CEMIn).

Outage times from meter last-gasp and restore messages (99.8% of meters are smart), checked against OMS.

Deliverable: tested views and a data dictionary

Days 61–90

Build the tracking report

Track mainfeeder hardening against matched untreated feeders (same pre-period performance), weather-normalized, in support of the program's selection rule.

Deliverable: a Power BI program-tracking report, refreshed on a schedule

Why it matters: these are questions to check first with the people who own each definition, not conclusions.

Source: A plan: Dominion data is named by type only, and none was used here. Smart meters: EIA-861 2025 (advanced meters as a share of all meters). Grid Resiliency: Dominion Energy website, read November 1, 2026.

Three takeaways: a stormier year, a frequency gap, and storms by county

1 **A stormier year**
More storms and more interruptions in 2025; no clear trend once storms are counted

2 **Frequency is the peer gap**
So causes are the question, and answering it needs outage records by cause code

3 **Storms by county**
Report reliability beside county storm exposure, so a stormy year isn't read as a trend

I've built a tested Snowflake → Power BI pipeline on public reliability data.

How my experience maps Mercury Insurance, Data Analyst

Snowflake pipeline, 20M+ rows
→ this project's mart

Root-cause analysis on joined data
→ what I'd do with cause codes in the first 90 days

Metadata and business definitions
→ the data dictionary

Thank you. I'd like to do this work with your team.

Source: Takeaways: slides 2–4. Experience: Mercury Insurance, Data Analyst, 2026 to present.

Definitions: SAIDI = SAIFI × CAIDI, and major event days are set aside by rule

SAIDI
outage minutes per customer per year

=

SAIFI
interruptions per customer per year: how often

×

CAIDI
minutes per interruption: how long

Dominion 2025: 1.320 × 109.4 = 144.4 minutes (excluding major event days)

Major event days (MED): IEEE 1366's 2.5 beta method takes five years of daily SAIDI, drops zero days, and sets α and β as the mean and standard deviation of ln(daily SAIDI); a day above exp(α + 2.5β) is an MED.

Dominion reports to EIA under its own major-event rule (“Other” in EIA-861); methods differ between utilities.

With major event days, 2025 SAIDI was 306.7 minutes, against 144.4 without them: MEDs held 53%.

Log split: ln(SAIDI ratio) = ln(SAIFI ratio) + ln(CAIDI ratio). From 2022, duration fell, so frequency accounts for more than all of the rise; from 2023, 79%.

Source: IEEE Std 1366; EIA-861 reliability, Dominion Energy Virginia, 2022, 2023 and 2025.

The weather fit is a check on one year, not a forecast

Year	Storm county-days	SAIDI ex-MED	From the fit
2015	526	119.6	131.4
2016	597	138.2	135.0
2017	420	117.4	125.9
2018	687	149.6	139.7
2019	851	149.9	148.1
2020	431	141.0	126.5
2021	676	133.6	139.1
2022	723	136.0	141.5
2023	391	123.3	124.4
2024	465	131.3	128.2
2025 (not in fit)	**796**	**144.4**	**145.3**

Fit on 2015–2024 (n = 10): SAIDI = 104.4 + 0.0514 × storm county-days; r = 0.70, R^2 0.48.

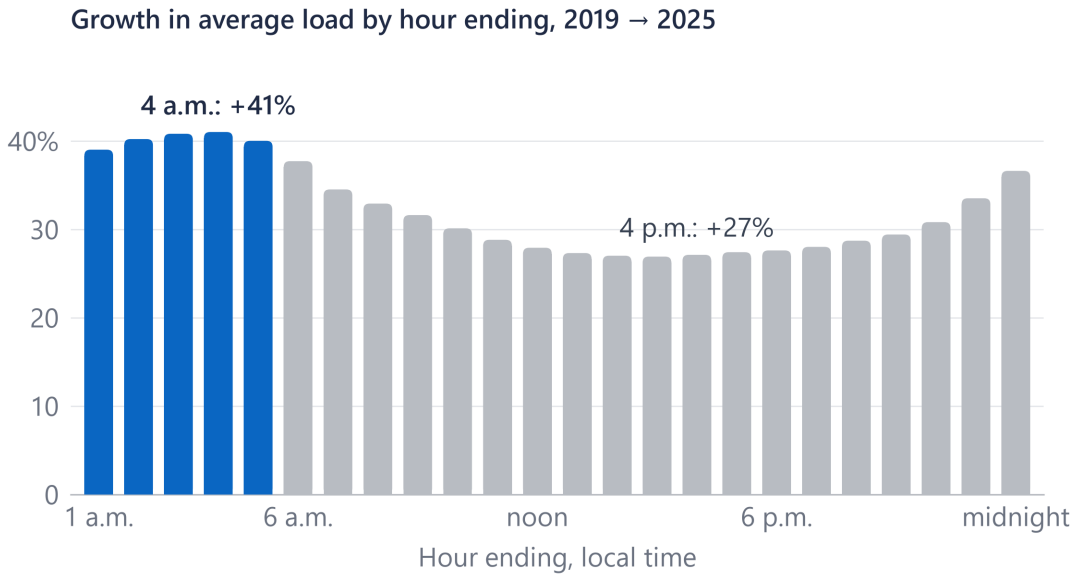
Why 2015: the change to automated outage recording that year shifts the level (decision D6).

2025 was left out: the fit gives 145.3, the actual was 144.4.

Adjusted for storm days, there is no clear trend since 2015.

Source: NOAA Storm Events and EIA-861, 2015–2025; ordinary least squares on ten annual points (exploratory).

Overnight demand grew faster than afternoon demand; winter rivals summer for the peak



+41%

average load in the hour ending 4 a.m., 2019 → 2025 (hour ending 4 p.m.: +27%)

Winter

set the yearly peak in 2019, 2022 and 2025

Highest hour of the year +26%, lowest hour +52%.

I'd add winter and overnight views to load and outage reports.

Source: EIA-930 hourly demand, PJM Dominion (DOM) zone, which is wider than Dominion's retail territory; 2019 and 2025 full years. Winter = Dec–Feb, summer = Jun–Aug.

Every number traces to a public source; the tables behind them are tested after each load

Source	Publisher	Years	Grain	Rows
EIA-861 Reliability, 2025 final	EIA	2025	utility × state × year	977
PUDL EIA-861 reliability	Catalyst Cooperative	2013–2024	utility × state × year	24,358
EIA-861 Service Territory	EIA	2025	utility × county	11,803
NOAA Storm Events	NOAA NCEI	2013–2025	storm event	831,288
EIA-930 hourly demand	EIA	Jul 2018–Oct 2026	region × hour	6,077,232
Dominion Energy 2025 Annual Report	Dominion Energy	2025	document (130, 133)	reference

Tests after each load (T1–T12, all pass): no duplicate keys; SAIDI = SAIFI × CAIDI within 2%; Dominion 2025 equals EIA's published 306.7, 144.4 and 2,675,735 customers; every county mapped; full years; no duplicate hours.

Source: Project documentation: sources register (publisher, vintage, license) and data-quality tests; all public data.

What public data cannot show, and what would answer it

Public data limit	What would answer it (Dominion data, by type)
One number per utility per year (EIA-861)	Outage records by event, circuit and cause
Storms counted by county, not by line	Outage records with cause codes and circuit location
Methods differ between utilities	Dominion's daily SAIDI and its filing method
The DOM zone is wider than the retail territory	Meter interval data by circuit
No customer identities: growth can't be attributed	Customer class and large-load records

Source: Project scope limits (decision D14) and the sources above.

The 12-utility comparison (Dominion and 11 peers), with each utility's reporting method

Utility	Major-event method	SAIFI 2021–25	CAIDI 2021–25	SAIDI 2023–25
Duke Carolinas (NC)	IEEE 1366	1.25	139	179
Georgia Power	Mixed	1.24	114	141
Duke Progress (NC)	IEEE 1366	1.23	114	144
Dominion Energy Virginia	**Own method**	**1.21**	**110**	**133**
Dominion Energy SC	IEEE 1366	1.14	72	84
Duke Energy Florida	Mixed	0.99	87	83
Alabama Power	Own method	0.93	125	110
PECO	IEEE 1366	0.74	99	66
BGE	IEEE 1366	0.71	99	64
PPL Electric	IEEE 1366	0.69	137	99
Pepco (MD)	IEEE 1366	0.57	90	48
FPL	IEEE 1366	0.49	96	47

Source: EIA-861 / PUDL, excluding major event days; means of annual values; sorted by SAIFI. Peers chosen by EIA ID before results (D10). Methods as labeled in the project's Power BI data; Mixed: “Other” in 2013 only, IEEE 1366 for 2014–25.